

02 - eFile Transmission Module

Overview

The eFile Transmission module is the unified orchestration layer that manages the full lifecycle of electronic return submission from initial staging through IRS acknowledgment and client notification.

Transmission Lifecycle

[Preparer Stages Return]

- > [Validation Engine: Schema + Business Rules]
 - > [ERO Signature Applied (PIN)]
 - > [Encryption + EFIN Header Injection]
 - > [Routed to Correct Channel]
 - Individual/Business -> MeF A2A
 - Information Returns -> FIRE
 - State Returns -> State Gateway
 - > [IRS Acknowledgment Received]
 - > [Client and Staff Notification]
 - > [Acknowledgment Stored in Masterfile]
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Validation Rules

Pre-Transmission Checks

Check	Rule	On Failure
SSN/ITIN Format	9 digits, no dashes	Block
EIN Format	XX-XXXXXXX	Block
EFIN Present	Must match 748335	Block
ERO PIN	5-digit self-select	Block
Duplicate Detection	Check prior submissions	Warn
Balance Due	IRS payment routing valid	Warn
Schema Version	Must match current MeF schema	Block
State Attachment	Required if state return present	Warn

Transmission Status Codes

Code	Description	Action Required
STAGED	Return queued, not yet sent	None
TRANSMITTING	In transit to IRS	Monitor
ACCEPTED	IRS accepted return	Notify client
REJECTED	IRS rejected with error codes	Correct and resubmit
PENDING	IRS processing, no ack yet	Wait up to 48 hours
DUPLICATE	Previously accepted return	Investigate
HOLD	ERO hold applied	Release or discard

Rejection Handling Workflow

```
[Rejection Received from IRS]
-> [Parse Rejection Code(s)]
-> [Map to Internal Error Description]
-> [Route to Preparer Queue with Fix Instructions]
-> [Preparer Corrects Return]
-> [Re-Validate]
-> [Resubmit: Max 3 attempts before escalation]
```

Common Rejection Codes

IRS Code	Description	Common Fix
R0000-504	Duplicate SSN on return	Verify SSN, check prior filing
R0000-902	EFIN not active	Verify IRS e-Services
IND-031	Prior year AGI mismatch	Correct AGI from prior return
IND-032	Prior year AGI mismatch (spouse)	Correct spouse AGI
F1040-072	Invalid IP PIN	Obtain correct IP PIN from client
R0000-507	SSN locked by identity theft	Client must contact IRS
DCN-DuplicateW2	Duplicate W-2 submitted	Consolidate or remove duplicate

Configuration

```
efile_transmission:  
  efin: "748335"  
  ero_pin_required: true  
  max_batch_size: 100  
  retry_attempts: 3  
  retry_backoff_seconds: [30, 60, 120]  
  duplicate_window_days: 365  
  acknowledgment_poll_interval_seconds: 90  
  channels:  
    federal: mef_a2a  
    information_returns: fire  
    state: state_gateway  
  notifications:  
    on_accept: [email, sms, in_app]  
    on_reject: [email, in_app]  
    on_pending_48h: [in_app, alert]
```

Audit Trail Fields

Every transmission event logs:

- transmission_id (UUID)
- efin (748335)
- timestamp_utc
- return_type (1040, 1120, 1065, and others)
- taxpayer_tin_masked (XXX-XX-1234)
- channel (mef_a2a | fire | state)
- status
- ack_code
- preparer_id
- ip_address